



A typical home electrical circuit with power saver

with the electricity meter. A power saver is connected to the electrical circuit after the meter, which means it does not hamper or manipulate meter functioning.

Can I use it in my house to save electricity?

Power savers can be used in flats, houses, shops or small offices. But power savings depend on the type of the load being used in the building.

Will it give me 30 per cent power savings as claimed by the manufacturer?

Most companies claim that their devices can save 30 to 40 per cent of power. However, the actual savings are only 5-10 per cent.

What are the parameters on which the savings depend?

The power savings depend on:

Type of house. In old houses, wear and tear in copper wires may result in current leakage, thus adding to the power loss. In such houses, power savers may facilitate better savings. However, in houses with new wiring, the percentage saving from power savers is relatively less.

Type of appliances. More inductive appliances in a house means more

TABLE I
Resistive and Inductive Loads

Type of load	Power saver benefit	Examples
Resistive	No	Bulb, geyser, electric iron
Inductive	Yes	Air-conditioner, refrigerator, water pump, washing machine, tubelight

TABLE II
Power Consumption by Different Devices

Device	Wattage	Number of units	Total wattage
Air-conditioner	1.5 kW	1	1.5 kW
Tubelight	40W	4	160W
...	...	...	...
Total			2.1 kW

power savings. The saving is only in terms of wastage or reactive power. There is no reduction in true power (actual working power) required by the appliances to function.

Ageing of appliances. Efficiency of any electronic/electrical appliance decreases with ageing. This also results in more heat losses. Power savers show significant savings with old appliances.

Locality. Areas with high voltage fluctuations like shops and restaurants can gain additional savings from power savers.

Electricity meter. It also affects the working of power savers.

Atmosphere. Working of air-conditioners and refrigerators depends on the atmospheric temperature. Thus the atmosphere plays an important role in power savings.

What is the role of electricity meter?

Electricity is not stable, as fluctuations occur from the supply side. At the same time, the load in houses is not balanced. Meters installed in houses are either traditional torque-operated

or electronic meters.

The traditional torque-operated meters are highly sensitive to spikes, giving incorrect readings sometimes. Electronic meters, on the other hand, are more accurate. All these factors influence the savings from power savers.

Visit 'www.derc.gov.in/ElectronicMeters/Awareness/FAQs.html#a1' to learn more about meter readings.

Under what conditions is the power saver really effective?

If you have only resistive load (bulbs, iron, geyser, etc), connecting a power saver to the electrical circuit won't give you any significant power saving. Even if you add a tubelight and a fan to these devices, you won't be able to save much. You can save only on those devices which contribute to the inductive load, such as refrigerators, washing machines, motors and air-conditioners.

Again, the percentage saving depends on the parameters mentioned earlier. Power savers save only the power that is wasted or is not useful.

Are power savers tested or certified?

Power saver devices may not necessarily be tested. However, agencies like National Test House perform tests and issue certificates. These tests do not comment on percentage saving. They provide only factual readings with or without the power saver in constant-load condition.

EFY Lab reviewed the test certificate of a power saver, which indicated six per cent savings for a window air-conditioner through the power saver.

What are the different types of power savers?

Power saver devices come in different kilo-watt (kW) ratings of power capacity. Commonly available are 3 kW, 6 kW, 10 kW and 15 kW. Gener-

Before choosing a power saver, you need to figure out the resistive and inductive loads in your house. You must have some inductive appliances to avail the power saving benefit